The Norns III: Seeding Simulated-Annealing Placement with Provenance and Recovered Clusters How hot can the anneal start before it destroys a good initial placement?

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***Abstract*—Simulated annealing is the workhorse of FPGA placement, and it starts from an initial placement that the anneal then destroys and rebuilds. What if the initial placement carried real information – the HDL hierarchy a provenance-tracked front end kept, or the clusters a graph analysis recovered from the flat netlist? This paper asks whether such seeds improve placement, and, because the anneal's early phase can erase any seed, at what initial temperature the seed's advantage dies. The seeds come from Odin III's provenance hierarchy and from Loki III's L2 clusterings (Louvain, spectral, affinity propagation, k-means); the die-level arrangement of clusters comes from a cut-weighted quotient-graph layout; and the seeds enter VPR through its initial-placement and floorplanning-constraint interfaces, with VPR's stock checkpointing flag (on by default, following RLPlace) enabled throughout and an off sweep as the control arm. We report seed retention at scale 1 and at twice each design's crossover, quality of results against an unseeded baseline, and the largest tested temperature scale at which each seed still helps – the recipe a clustering-based initial placement should be used with.**

***Index Terms*—FPGA placement, simulated annealing, provenance, clustering, VTR**

I. Introduction

VPR's simulated-annealing placer starts from an initial placement and anneals it: the schedule begins hot enough to move anything, and the early phase rearranges the circuit wholesale. The question this paper asks is whether the initial placement can be made to *mean* something, and how much of that meaning survives the anneal. Gharibian et al. showed that system-level structure recovered from a flat netlist correlates with placement locality [1]–[3]. Odin III keeps the provenance link that makes recovery unnecessary for the hierarchy [1], and Loki III recovers clusters from the flat netlist without a language model. RLPlace [4] showed that the anneal itself can be improved – directed moves chosen by a reinforcement-learning agent and checkpointing that prevents oscillation – starting from VTR's analytic placement. This paper connects those lines: the seed carries the structure, the anneal is measured for how much of it the seed can keep, and the largest tested temperature scale at which the advantage survives is reported as data, not tuned away.

The contributions are: (1) a seeding methodology – provenance hierarchy and four recovered clusterings, arranged on the die by a quotient-graph layout, entering VPR through its initial-placement and constraint interfaces, with VPR's stock checkpointing enabled and its off sweep as the control arm; (2) the crossover sweep, which calibrates the highest useful initial temperature per design against the best placement any seed achieves; (3) the retention and quality numbers per seed, with the full-seed versus inter-cluster-only split, which measures sensitivity to the intra-cluster order; the coarse-versus-fine question remains open.

II. Background

A. Simulated annealing placement and RLPlace

VPR places by simulated annealing from an analytic initial placement [5], [6]. The anneal's moves are mostly random; its early phase can undo any structure the start carried. RLPlace improves the anneal rather than the start: a K-armed bandit selects among directed move types (simple moves and swaps, bounded centroid, median, critical-path moves), and checkpointing periodically saves the best placement so oscillation cannot throw it away [4]. Our experiment keeps the anneal as-is (VPR already ships RLPlace-style checkpointing, on by default) and changes the start instead; the checkpointing prediction – that save-best/restore extends the useful temperature range – is one of the hypotheses we test.

B. The seeds

The provenance hierarchy (Urd) is the module structure Odin III records on every netlist object; it needs no recovery. Urd has a single cluster only on sha and stereovision3, two on diffeq1 and diffeq2 (their one module plus a flat group), and one per populated module (3 to 23) on the other designs, so its signal is degenerate on the two one-cluster designs. The recovered clusters (Verdandi) are Loki III L2 partitions: Louvain, spectral, affinity propagation and k-means at several k, scored in the Loki III paper [7] against the known hierarchy. Agent default: Norns builds the Loki L0 graph.json from the same Odin III BLIF, because Loki's key.json uses Odin API

names that do not join to VPR atom names; the clustering code is Loki's. The two seed families differ exactly in the question's terms: provenance is the structure the designer wrote, clusters are the structure the flat netlist still shows.

III. Methodology

A. Seed construction

Each seed is a partition of the placed blocks. Its die-level arrangement comes from the quotient graph: contract each cluster to a node, weight edges by inter-cluster net cuts, lay the quotient out by a coarse force-directed solve, and scale it to the die. Inside a cluster, blocks take the seed's internal order, or a shuffle in the inter-cluster-only arm; the two arms differ only in that intra-cluster order, not in the coarse cluster layout.

B. Injection

Seeds enter VPR two ways, both already in the pinned build (v8.0.0+18416): soft, as an initial placement file (`--read_initial_place_file`) that the anneal is free to move away from; and hard, as floorplanning constraints (`--read_vpr_constraints`) with one PartitionRegion per cluster, which the anneal cannot violate at any temperature. The hard arm constrains the regions only, so the anneal chooses the fine placement inside each one (one hard variant, no full/inter split). `--write_initial_place_file` dumps VPR's own analytic start for the baseline study. VPR's placement macros are respected: I/O pads and the multiplier tiles stay anchored, and carry chains move as one vertical run. The seeds are built from an Odin III provenance BLIF and Loki III's clusterings of the same netlist, arranged by a cut-weighted quotient-graph layout with `networkx` (spring layout, fixed seed). Building the seed files also needs VPR's unseeded baseline placement for legal slots, so each design's baseline run precedes its seeds.

C. The crossover sweep

The calibration loop follows the data, in order. First, run every seed and keep the best final placement P^* (and which seed made it). Second, cold-start a fresh anneal from P^* and sweep the initial temperature upward; the crossover is the largest tested scale at which the cold start still routes shorter than the unseeded run at the same scale, a lower bound on the true fall point when the sweep stops at its cap. The sweep is run once per design, so S is design-level and identical across that design's seeds. The temperature is reported as a multiple S of VPR's automatic start-temperature estimate. Third, run every seed at twice the design's crossover against an unseeded run at the same temperature: whether the seed still wins there tests how much of its advantage survives a hot restart. The checkpointing arm repeats the sweeps with VPR's checkpointing disabled; per RLPlace, the crossover should move hotter when it is on.

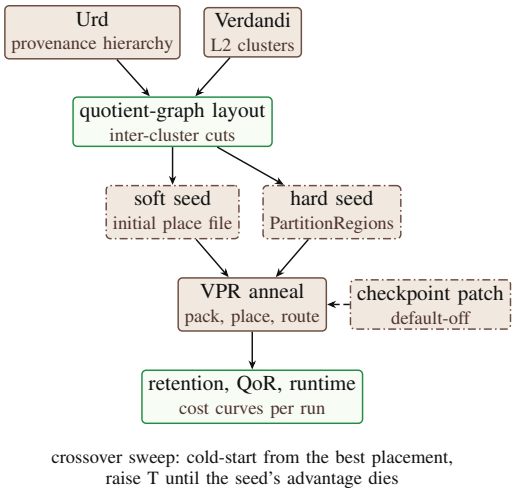

Fig. 1. The Norns III experiment. Seeds from provenance and recovered clusters, arranged by a quotient-graph layout, enter VPR softly (initial placement) or hard (constraints). The crossover sweep cold-starts from the best placement and raises the initial temperature until the seed's advantage dies or the sweep reaches its cap; the checkpointing arm (VPR's stock flag, on vs off) tests whether save-best/restore extends the range.

TABLE I
GEOMEAN QUALITY RATIOS VS THE UNSEEDED BASELINE, PER SEED FAMILY, FOR THE SOFT INITIAL-PLACEMENT ARM; THE INTER-CLUSTER-ONLY ROW SHUFFLES THE ORDER INSIDE EACH CLUSTER AND POOLS ALL SEVEN SEED FAMILIES, AND THE K-MEANS ROW POOLS k, $2k$ AND $4k$. BELOW 1.0 IS BETTER. THE HARD-CONSTRAINT ARM IS DISCUSSED IN THE TEXT.

Seed	Wirelength	Delay	Chan. width	Runtime
Urd (hierarchy)	1.029	1.004	0.958	1.068
Louvain	1.008	0.982	0.980	1.091
Spectral	1.006	0.993	0.968	1.098
Affinity propagation	1.002	1.003	0.982	1.086
k-means (k, $2k$, $4k$)	1.006	0.985	0.975	1.088
Inter-cluster only	1.015	0.981	0.987	1.102

D. Metrics and data

Per run: seed retention (fraction of cluster CLBs still inside their seed cluster's region at exit; mean displacement from the seed coordinate), routed wirelength, critical-path delay, minimum channel width, placement runtime, and the per-temperature cost curve. Designs are the 15 VTR benchmarks that fit the development machine, smallest first, one run at a time under the heavy lock. Every number in this paper traces to a kept record; large data keeps its scripts.

IV. Results

The 15 completed designs give the ratios of Table I. Soft seeds stay within a few percent of the unseeded baseline in routed wirelength (1.009 for the full seed, 1.015 for the inter-cluster-only arm, both pooled over all seven seed families) while cutting the minimum channel width to 0.973 and costing 1.087 in placement runtime (both also pooled). The hard arm is worse on average (geomean 1.088 over the 104 runs that completed; 14 of those still beat the baseline, and stereovision3's hard geomean is 0.906), but not uniformly: one

TABLE II

THE CROSSOVER SWEEP, PER SEED FAMILY: THE LARGEST TESTED SCALE S OF VPR'S AUTOMATIC START-TEMPERATURE ESTIMATE AT WHICH THE COLD START FROM P^* (THE BEST PLACEMENT OVER ALL SEEDS) STILL ROUTES SHORTER THAN AN UNSEEDED RUN AT THE SAME S, AND THE SEED'S MEDIAN RETENTION AT THE HOT STEP ($2S$). S IS DESIGN-LEVEL (ONE SWEEP PER DESIGN), SO IT IS IDENTICAL ACROSS FAMILIES; WHERE THE SWEEP STOPPED AT ITS CAP, S IS A LOWER BOUND.

Seed	Crossover S	Retention
Urd (hierarchy)	16×	0.744
Louvain	16×	0.520
Spectral	16×	0.635
Affinity propagation	16×	0.564
k-means (k, $2k$, $4k$)	16×	0.667

blob_merge k-means$_{2k}$ run is missing from the arm because VPR's initial placement could not fit 9 CLBs inside its regions (the constraints are too tight, not a packing failure). Fixed regions may block the anneal's fine moves across cluster borders – a hypothesis the aggregate numbers are consistent with. No single seed family dominates: Urd is the worst full seed in soft wirelength (1.029) and the best hard seed (1.062 against 1.075-1.120 for the recovered families). Retention at scale 1 is 0.68 of a seed's cluster CLBs (full) and 0.70 (inter).

V. DISCUSSION

Three observations. First, the seed families span a narrow band but are not interchangeable: provenance (Urd) is the worst full seed in the soft arm (1.029) and the best in the hard arm, so it is not consistently better or worse than recovered clusters. The full and inter-cluster-only arms differ by 0.6% in pooled wirelength (1.009 vs 1.015); no arm randomizes the coarse cluster layout, so whether the fine order inside a cluster matters as much as the coarse die position is not measured here and stays open. Second, VPR's checkpointing does not separate the crossover in the measured range: the two sweeps find the same scale on 14 of the 15 designs where both found one, and the control extends it on stereovision0 (control 32 vs 16). RLPlace's prediction that checkpointing extends the useful range is therefore not observed on these designs; if it holds, its effect is smaller than the between-design variation. Third, the safe recipe is to cold-start from a cluster-derived placement at a temperature scale at or below the crossover, which is a multiple of VPR's own estimate; above it the seed's retention is still high per family (medians 0.52-0.74 at twice the crossover, individual runs 0.21-1.00) but it wins only 57 of 105 head-to-heads against an unseeded run at the same temperature – the coarse structure survives the anneal, the advantage does not.

VI. CONCLUSION

Seeding VPR's anneal is worth it only as a calibrated cold start: at VPR's own start temperature the cluster layouts and the provenance hierarchy place at parity with the analytic start, but from a good final placement they keep winning when the anneal is restarted hotter, up to the largest tested scale that still wins – a lower bound on the fall point where the sweep hit its cap. The recipe is that crossover table, not a seed-specific tuning: start at or below it, and expect the seed to inform the coarse placement while the anneal supplies the fine one.

REFERENCES

[1] F. Gharibian, L. Shannon, and P. Jamieson, "Finding system-level information and analyzing its correlation to FPGA placement," in *Proc. Int. Conf. Field Programmable Logic and Applications (FPL)*, 2010, pp. 544–549.

[2] F. Gharibian, L. Shannon, P. Jamieson, and K. Chung, "Analyzing system-level information's correlation to FPGA placement," *ACM Trans. Reconfigurable Technology and Systems*, vol. 6, no. 3, pp. 15:1–15:21, 2013.

[3] F. Gharibian, L. Shannon, and P. Jamieson, "Identifying and placing heterogeneously-sized cluster groupings based on fpga placement data," in *Field Programmable Logic and Applications , 2014 24th International Conference on*. IEEE, 2014, pp. 1–6.

[4] M. A. Elgammal, K. E. Murray, and V. Betz, "RLPlace: Using reinforcement learning and smart perturbations to optimize FPGA placement," *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, vol. 41, no. 8, pp. 2532–2545, Aug. 2022.

[5] J. Luu, J. Goeders, M. Wainberg, A. Somerville, T. Yu, K. Nasartschuk, M. Nasr, S. Wang, T. Liu, N. Ahmed, K. B. Kent, J. Anderson, J. Rose, and V. Betz, "VTR 7.0: Next generation architecture and CAD system for FPGAs," *ACM Trans. Reconfigurable Technology and Systems*, vol. 7, no. 2, pp. 6:1–6:30, Jun. 2014.

[6] K. E. Murray, O. Petelin, S. Zhong *et al.*, "VTR 8: High-performance CAD and customizable FPGA architecture modelling," *ACM Trans. Reconfigurable Technology and Systems*, vol. 13, no. 2, 2020, article 9.

[7] P. Jamieson, "Loki III: Reverse engineering structural netlists without a language model," 2026, manuscript.